

UNIT 4

DATA AND ANALYSIS

9TH COMPUTER SCIENCE FEDERAL BOARD

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AGENDA

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DATA

- Data is an information or facts that we gather about something. It can be represented by numbers, measurements, descriptions, sounds or pictures.
- Here are two examples of data. In a science experiment when you record the temperatures at different times, that temperature values are data. If you conduct a survey of your classmates and get to know how many of them like Mathematics, it will be called data.

DATA ANALYTICS

- Data Analytics refers to the process of carefully studying data to identify patterns and draw conclusions.
- To analyze data, we use mathematical calculations, statistical techniques, charts, or other tools to understand data.
- For example, after recording hourly temperature data, you can create a graph to see how it changed over time. From the graph, you conclude that it got warmer as the day went on — that is the result of your data analysis.
- So, data is the information we collect, and data analytics is how we observe it to get meaningful information. Data analytics can be quantitative (numeric – height, temperature, marks) or qualitative (descriptive - color, gender, taste)

DATA SCIENCE

- Data Science refers to an interdisciplinary field of mathematics, statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it.
- The insight result drawn can be used for making informed decisions to solve real world problems e.g., medical, education, scientific research, and business etc.

CONCEPTS OF DATA SCIENCE

Data

A data is an information collected from different sources. It could be structured in the form of tables or unstructured in the form of audio, video, tweets, pdf files etc.

Dataset

A collection of related data organized in a structured way which is ready for the project of data science.

Statistics and Probability

Statistics is the analysis of the frequency of past events and probability is to predict the likelihood of future events. Data scientists use statistics and probability to find patterns and trends in the data.

CONCEPTS OF DATA SCIENCE

Mathematics	Mathematics is a fundamental part of data science which helps to solve problems, optimize the model performances, and interpret huge complex data into simple and clear results, for decision making.
Machine Learning	Machine Learning is a branch of Artificial Intelligence (AI) that allows computers to learn from data without being explicitly programmed.
Deep Learning	Deep learning is the subset of Machine learning, with emphasis on the simulation or imitation of human brain's behavior by using artificial neural networks.

CONCEPTS OF DATA SCIENCE

Data Mining	Data mining is the process of discovering patterns and relationships in existing datasets. The usage of techniques and tools is limited in data mining as compared to data science.
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Data Visualization	Data visualization is the graphical representation of data using common charts, plots, infographics, and animations.
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Big Data	Big data refers to handling large volumes of data. The huge size of data provides more opportunities for machine learning and provides better results.
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CONCEPTS OF DATA SCIENCE

Predictive Analysis

Predictive analysis is the use of data to predict future trends and events based on historical data.

Natural Language Processing (NLP)

It is the study of interaction between human language and computers. The common uses of NLP are chatbots, language translators and sentiment analysis.
